

The Variable-Force Root Modal in Laz is Gradable¹

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Abstract. This study documents the syntactic and semantic properties of the modal suffix *-oni* in Laz, which contributes a root modal that exhibits force variability. The verbal suffix *-oni* constructs nonfinite predicates that serve both attributive and predicative functions in the language, akin to infinitivals in English. We argue that it denotes a gradable modal, as evidenced by its ability to participate in comparative structures. The degree-based account in Grano (2024) we adopt for *-oni* not only explains its force variability but also accounts for how different syntactic contexts restrict it, in line with the Determiner-Modal Generalization (Bhatt, 1999; Hackl and Nissenbaum, 2012).

1 Introduction

Laz has a verbal suffix *-oni* that builds a nonfinite predicate akin to infinitivals in English (Bhatt, 1999), as illustrated in (1).² We refer to these nonfinite clauses as ONI-clauses, which have both attributive and predicative uses.

- (1) a. *Ham çoyi-s, [o-nç'val-oni] pucepe on.*
 this village-LOC PV-milk-ONI cow.PL be.PRS.3SG
 ‘In this village, there are cows to be milked.’
 i.e. There are cows that **can/should** be milked. Attributive ONI-clause
- b. *Ham puci [o-nç'val-oni] on.*
 this cow PV-milk-ONI be.PRS.3SG
 ‘This cow is to be milked.’
 i.e. This cow **could/should** be milked. Predicative ONI-clause

We make a number of observations about the syntax and semantics of ONI-clauses. First, ONI-clauses systematically contribute a root modal, which exhibits force variability (Bhatt, 1999; Deal, 2011; Bochnak, 2015; Rullmann et al., 2008; Peterson, 2010; Kratzer, 2012; Jeretič, 2021). It is noteworthy that its force variability survives under clause-mate negation, as shown in (2).³

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²Laz is an endangered language spoken primarily in Turkey, belonging to the South Caucasian/Kartvelian language family. Speakers are bilinguals in Turkish, with varying degrees of proficiency in Laz (Avcı-Bucaklışı et al., 2018; Eren, 2023). This study reports data from Pazar (Atina) dialect of Laz (Öztürk and Pöchtrager, 2011; Bucaklışı and Kojima, 2003).

³This is in contrast with the variable-force ability-compulsion modal in Laz, which only allows the $\neg\Diamond$ reading under negation (Demirok, 2023).

- (2) a. *Ma [oxori gama-çam-oni] vore.*
 I house PV-sell-ONI be.PRS.1SG
 ‘I could/should sell my house.’
- b. *Ma [oxori gama-çam-oni] var vore.*
 I house PV-sell-ONI NEG be.PRS.1SG
 ‘I cannot sell my house.’ or ‘It’s not the case that I should sell my house.’

Nevertheless, there are also syntactic contexts that bar one of the two interpretations. We show that this behavior of ONI-clauses falls under the DETERMINER-MODAL GENERALIZATION put forth in Hackl and Nissenbaum (2012) for *for*-infinitival relatives in English (Bhatt, 1999), according to which there is a systematic interaction between the force of the modal and whether the determiner is strong or weak.

- (3) a. *K’at’a gama-çam-oni oxori ham çoyi-s on.*
 every PV-sell-ONI house this village-LOC be.PRS.3SG
 ‘Every house that should/#can be sold is in this village.’
- b. *Ham çoyi-s (bazi) gama-çam-oni oxorepe on.*
 this village-LOC (some) PV-sell-ONI house.PL be.PRS.3SG
 ‘In this village, there are (some) houses that should/can be sold.’

Second, there is robust evidence that ONI-clauses denote gradable predicates. They are able to serve as gradable predicates in comparative structures, as illustrated in (4). Accordingly, in analyzing ONI-clauses, we will adopt the degree-based semantics for modality defended by Grano (2024), arguably offering the first viable explanation of the DETERMINER-MODAL GENERALIZATION mentioned above.

- (4) *Şk’imi oxori sk’ani oxori-şa daa gama-çam-oni on.*
 my house your house-ABL more PV-sell-ONI be.PRS.3SG
 ‘My house is more sellable than your house.’

Third, we observe that ONI-clauses in attributive position cannot modify NPs that would be construed as an agent of the event, as shown in (5). On the other hand, in predicative position, ONI-clauses exhibit no such restriction, as was shown by the examples in (2). We will provide a compositional treatment of this asymmetry, arguing that the syntactic complement of ONI never contains a voice layer (Kratzer, 1996).

- (5) a. *[e₁ gama-çam-oni] a oxori₁*
 PV-sell-ONI a house
 ‘a house that can/should be sold’
- b. **[oxori gama-çam-oni] a k’oçi*
 house PV-sell-ONI a man
 Int. ‘a man that can/should sell the house’

The paper is structured as follows: Section 2 gives an overview of ONI-clauses and establishes it as a root modal with force variability. Section 3 takes a closer look at the factors concerning the distribution of the force variability and situates the contribution of ONI within the approach proposed in Grano (2024). Section 4 investigates the structural makeup of attributive and predicative ONI-clauses and provides compositional analyses for them. Section 5 concludes with a brief summary.

2 An overview of ONI as a modal

In this section, we argue that ONI is a suffix that builds nonfinite verbal predicates. Unlike infinitives in English, they do not appear to have a function (such as serving as control clauses) other than introducing a root modal. We show which syntactic frames ONI-clauses can occur in. Finally, we justify our claim that ONI contributes a modal that exhibits force variability, showing its felicitous use in different contexts.

2.1 ONI-clauses are syntactically specialized nonfinite clauses

We refer to all clauses containing a verbal root marked by the invariant suffix *-oni* as an ONI-clause.⁴ On the predicate of an ONI-clause, there is always a preverb, which is by default *o-*. If the root co-occurs with one of the many spatial prefixes in the language (Öztürk and Eren, 2021), that prefix shows up instead of the default *o-*. This is illustrated in (6). ONI-clauses are not the only type of nonfinite clauses.⁵ For example, so-called masdar (nominal) forms of verbal roots, which can function as control clauses in the language, have a similar composition, requiring the *o-*preverb as the default if no other spatial marker is to occur with the root, as shown in (7).⁶

- | | |
|--|--|
| <p>(6) a. <i>o- k'vat -oni</i>
 PREVERB- cut -ONI
 ‘ONI + cut’</p> <p>b. <i>gama- çam -oni</i>
 PREVERB- sell -ONI
 ‘ONI + sell’</p> | <p>(7) a. <i>o- k'vat -u</i>
 PREVERB- cut -NMLZ
 ‘to cut, cutting’</p> <p>b. <i>gama- çam -u</i>
 PREVERB- sell -NMLZ
 ‘to sell, selling’</p> |
|--|--|

ONI-clauses are morphologically quite undecorated, compared to finite verbal forms, which can host tense, and aspect affixes, ϕ -agreement affixes (for subjects as well as objects), valency affixes (causative, impersonal, applicative, etc.), and so forth (Öztürk and Pöchtrager, 2011). An example of a finite verbal form is given in (8).

⁴See also Bucaklışı and Kojima (2003) and Öztürk and Pöchtrager (2011) for brief discussion on *-oni*.

⁵Others include the participles *k'vat-eri* ‘cut, by cutting’ and negative participles *u-k'vat-u* ‘uncut, not cut’ (Öztürk and Pöchtrager, 2011).

⁶On masdar and finite complementation strategies in Laz, see Emgin (2009); Emgin and Öztürk (2011); Demirok and Öztürk (2022, 2024).

- (8) *Bere-k puci m- i- nç'val -am -t' -u*
 child-ERG cow 1- APPL- milk -IMPF -AUX -PST.3SG
 'The child was milking the cow for me.'

Predicative ONI-clauses are structurally analogous to finite sentences with non-verbal predicates, as exemplified in (9).

- (9) a. *Xordza [nçai o-tzil-oni] ort'u.*
 woman tea PV-pick-PCTP be.PST.3SG
 'The woman was (supposed) to pick tea.'
- b. *Xordza oxori-s ort'u.*
 woman house-LOC be.PST.3SG
 'The woman was in the house.'

Both feature a *be* auxiliary, bearing tense and ϕ -agreement. as illustrated in (10).

- (10) a. *Si [o-çalış-oni] ore*
 you PV-work-ONI be.PRS.2SG
 'You should work.'
- b. *Si msk'va ore*
 you pretty be.PRS.2SG
 'You are pretty.'

As mentioned in the introduction, ONI-clauses can also have attributive uses where they modify an NP, understood as the theme/object of the ONI-clause.

- (11) [*e₁ gama-çam-oni*] *ar oxori₁*
 PV-sell-ONI a house
 'a house that can/should be sold'

Besides these two main syntactic frames, ONI-clauses also frequently occur with the main verb 'become'.

- (12) a. *Şk'u [o-k'vand-oni] de-v-i-y-i-t.*
 we PV-beg-ONI PV-1-REFL-become-PST.1-PL
 'We came to the point where we can/should beg.'
- b. *Ham k'oçi [o-p'il-oni] do-m+a-y-u.*
 this man PV-shoot-ONI PV-1+APPL-become-PST.3SG
 Lit. 'This man became shoot-able for me.'
 (Context: He made me so angry that I could/should shoot him.)
- c. *Tomalepe [o-bğal-oni] do-m+a-y-u.*
 hair.PL PV-shave-ONI PV-1+APPL-become-PST.3SG
 'My hair became suitable for/in need of shaving.'

2.2 ONI is a prioritizing root modal

ONI is employed to make modal claims with teleological, bouletic, and deontic flavors, as illustrated in (13). Hence, it can be characterized as a prioritizing modal in the sense of Portner (2009).

- (13) a. *Ham bere [o-k'an-oni] on.*
 this child PV-rock-ONI be.PRS.3SG.
 'The baby should be rocked (to sleep).' TELEOLOGICAL
- b. *Si [o-ncir-oni] ore.*
 you PV-sleep-ONI be.PRS.2SG.
 'You should go to sleep.'
 (Context: uttered by a parent to a child) BOULETIC
- c. *Ham k'oçi [xapisi-s mola-xun-oni] on.*
 this man jail-LOC PV-enter-ONI be.PRS.3SG.
 'This man must go in jail.' DEONTIC

ONI is not used for expressing generic agentive abilities. However, it can express an agent's stage-level mood/will to do something.

- (14) a. *O-bir-oni vore.*
 PV-sing-ONI be.PRS.1SG
 Unavailable: 'I am able to sing.'
 Available 'I can sing (i.e. am in the mood to sing).'
- b. *Ma [çxombi o-şk'om-oni] vore.*
 I fish PV-eat-ONI be.PRS.1SG
 Unavailable: 'I am able to eat fish. (e.g. I am not allergic).'

ONI also cannot be used as an epistemic modal. Epistemic claims are made via adverbs that occur in finite clauses.

- (15) a. *Araba-muşı hak var on. #Noğa-şe o-lv-oni on.*
 car-his here NEG be.PRS.3SG. bazaar-ALL PV-go-ONI be.PRS.3SG
 Int. 'His car is not here. He must have gone to the bazaar.'
- b. *Araba-muşı hak var on. Noğa-şe idu abat'a.*
 car-his here NEG be.PRS.3SG. bazaar-ALL go.PST.3SG probably
 'His car is not here. He probably went to the bazaar.'

Further evidence that ONI is used to make modal claims comes from the absence of actuality entailments (Bhatt, 1999; Hacquard, 2020; Nadathur, 2020). As illustrated in (16), there is no entailment that the event actually took place, showing that ONI is a modal.

- (16) *Tzitzila o-ğur-oni vort'i ama va v-o-ğur-in-i.*
 snake PV-die-ONI be.PST.1SG but NEG 1-CAUS-die-CAUS-PST.1SG
 (i) 'I could kill the snake but I did not kill it.'
 (Context: I had a big stone in my hand.)
 (ii) 'I was supposed to kill the snake but I did not kill it.'
 (Context: The snake was poisonous.)

2.3 ONI is a variable-force modal

ONI exhibits force variability. As shown in (17), the very same sentence allows for two interpretations, depending on the context. If uttered in a context where the woman is wealthy and has money to buy a house, the sentence is understood as stating that the circumstances of the woman (e.g. her financial situation) would allow her to buy a new house. If uttered in another context where the woman's only house is old and damaged, the sentence is understood as stating that the circumstances of the woman (e.g. her living conditions) require her to buy a new house.

- (17) *Xordza [ağanı ar oxori o-indr-oni] on.*
 woman new a house PV-buy-ONI be.PRS.3SG
 (i) 'The woman **could** buy a new house.'
 (ii) 'The woman **should** buy a new house.'

Even without much context, the event being described seems to favor one interpretation over the other. Against the cultural context that paths in rural areas may or may not have barriers like bushes, branches on them, the two sentences in (18) clearly favor one interpretation over the other.

- (18) a. *Ham gzalepe o-lv-oni oran.*
 this path.PL PV-go-ONI be.PRS.3PL
 'These paths {**can** >> **should**} be passed through.'
 b. *Ham gzalepe go-klim-oni oran.*
 this path.PL PV-clean-ONI be.PRS.3PL
 'These paths {**should** >> **can**} be cleaned.'

An important finding about the force variability of ONI is that it survives under clause-mate negation, as shown by the pair of sentences in (19). The disambiguating sentences make it clear that distinct interpretations are accessed for the ONI-clauses under negation.

- (19) a. *Ma oxori gama-çam-oni var vore. Ar oxori xvala miğun.*
 I house PV-sell-ONI NEG be.PRS.1SG. a house only have.PRS.1SG
 'I can't sell (my) house. I only have one house.'
 b. *Ma oxori gama-çam-oni var vore. Para miğun.*
 I house PV-sell-ONI NEG be.PRS.1SG. money have.PRS.1SG
 'I don't have to sell (my) house. I have (enough) money.'

Next, we discuss syntactic factors restricting this baseline force variability of ONI-clauses, and show that ONI's force variability follows from ONI being a degree-based modal.

3 ONI as a gradable root modal

We adopt the approach proposed for force variability by Grano (2024), who builds on earlier work defending degree-based semantics in the analysis of modality (Portner and Rubinstein, 2016; Lassiter, 2011, 2017; Klecha, 2014). Grano (2024) sets out to solve the puzzle identified in Hackl and Nissenbaum (2012) for *for*-infinitival relatives in English (Bhatt, 1999). He argues that the covert modal that *for*-infinitival relatives contain is gradable, just like adjectives like *tall*. The force variability in *for*-infinitival relatives is tied to what ends up binding the degree variable that the modal brings in. The two interpretations are dubbed COULD and SHOULD readings.

- (20) There is a book for you to read.
 (i) There is a book [that you can/could read] on the table. COULD-reading
 (ii) There is a book [that you should read] on the table. SHOULD-reading

The SHOULD reading is analyzed akin to the positive uses of gradable adjectives via a null POS morpheme which has a denotation along the lines of (21). The COULD reading, on the other hand, is analyzed as default existential closure, only available within the VP domain in the sense of Diesing (1992).⁷

- (21) $\llbracket \text{POS} \rrbracket^c = \lambda P_{\langle d, t \rangle}. \text{MAX-d: } P(d) > d_c$
 where d_c is a contextually supplied standard on the relevant scale

Consider again the sentence in (22), for which (23) provides the derivations of the SHOULD and COULD readings, illustrating force variability is derived.

- (22) *Puci o-nç'val-oni on.*
 cow PV-milk-ONI be.PRS.3SG
 (i) 'The cow **could/can** be milked.'
 (ii) 'The cow **should** be milked.'

As a shorthand for now, we use $\mu_c(E)(d)$ to express: E is d-possible on the contextually provided measure function.⁸ (We return to the composition of ONI-clauses in Section 4).

⁷Grano (2024) argues that mere existential closure predicts triviality, and supplements existential closure with an 'above the minimum' condition. We're glossing over this technical detail here.

⁸Needless to say, this is not at all an explicit formalization. We refer the reader to Klecha (2014) and Grano (2024) in particular. See also the updated entry in (32).

- (23) a. MAX-d such that $\mu_c(\text{milk}(\text{the-cow}))(d) > d_c$
- $$\begin{array}{c}
 \text{puci} \quad \lambda_4 \quad \text{VP} \\
 \text{MAX-d such that } \mu_c(\text{milk}(\text{g}(4)))(d) > d_c \quad \text{V} \\
 \text{POS} \quad \lambda d. \mu_c(\text{milk}(\text{g}(4)))(d) \quad \text{on} \\
 \text{t}_4 \quad \lambda x. \lambda d. \mu_c(\text{milk}(x))(d) \\
 \text{on}\check{\varsigma}'\text{valoni}
 \end{array}$$
- b. $\exists d$ such that $\mu_c(\text{milk}(\text{the-cow}))(d)$
- $$\begin{array}{c}
 \text{puci} \quad \lambda_4 \quad \text{VP} \\
 \exists d \text{ such that } \mu_c(\text{milk}(\text{g}(4)))(d) \\
 \exists_{\text{closure}} \quad \text{V} \\
 | \quad \text{on} \\
 \lambda d. \mu_c(\text{milk}(\text{g}(4)))(d) \\
 \text{t}_4 \quad \lambda x. \lambda d. \mu_c(\text{milk}(x))(d) \\
 \text{on}\check{\varsigma}'\text{valoni}
 \end{array}$$

The proposal also correctly predicts that under clause-mate negation, POS and $\exists_{\text{closure}}$ will be available, accounting for the fact that the ambiguity survives under negation, as well.

- (24) *Puci o-n\check{\varsigma}'val-oni var on.*
 cow PV-milk-ONI NEG be.PRS.3SG
 (i) ‘The cow **cannot** be milked.’
 $\neg \exists d$ such that $\mu_c(\text{milk}(\text{the-cow}))(d)$
 (ii) ‘There is no need for the cow to be milked.’
 $\neg$ MAX-d such that $\mu_c(\text{milk}(\text{the-cow}))(d) > d_c$

As mentioned in the introduction, analyzing ONI as gradable modal predicts that it will be able to serve as a predicate within comparative structures. This is borne out, as we show below. A comparative sentence where the gradable predicate is a simple adjective is given in (25a). It seems that the morpheme *daa* which we gloss as ‘more’ could be given the denotation in (25b), predicting the truth conditions in (25c).

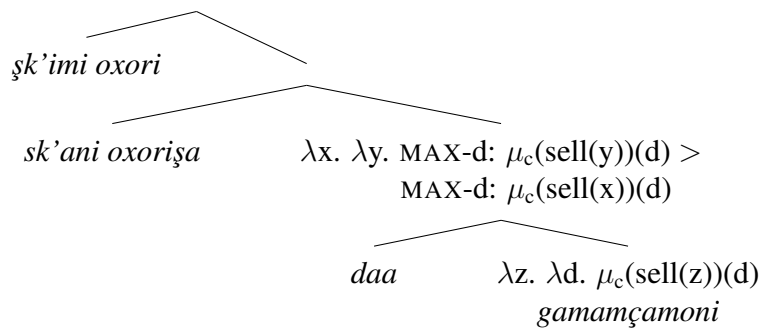
- (25) a. *Atina Artışeni-şā daa msk'va on*
 Atina Artışeni-ABL more pretty be.PRS.3SG
 'Atina is prettier than Artışeni.'
- b. $\llbracket daa \rrbracket = \lambda P_{\langle e, \langle d, t \rangle \rangle} \cdot \lambda x. \lambda y. \text{MAX-d: } P(y)(d) > \text{MAX-d: } P(x)(d)$
- c. $\llbracket \text{Atina Artışeni-şā daa msk'va on} \rrbracket = 1$ iff
 MAX-d: pretty(Atina)(d) > MAX-d: pretty(Artışeni)(d)

As shown by the examples in (26), ONI-clauses can function akin to as gradable adjectives, participating in the comparative structure.

- (26) a. *Şk'imi bere sk'ani bere-şā daa o-limb-oni on.*
 my child your child-ABL more PV-love-ONI be.PRS.3SG
 'My child is more lovable than your child.'
- b. *Ham lacepe doktori-k na nç'ar-u lacepe-şā daa*
 this medicine.PL doctor-ERG COMP write-PST.3SG medicine.PL-ABL more
o-şv-oni on.
 PV-drink-ONI be.PRS.3SG
 'These pills are easier to swallow than the pills that the doctor prescribed.'
 (Context: The ones that the doctor prescribed made me nauseous. The ones at home go down more easily.)

In (27), we sketch a derivation, using the denotation assumed for *daa*.

- (27) a. *Şk'imi oxori sk'ani oxori-şā daa gama-çam-oni on.*
 my house your house-ABL more PV-sell-ONI be.PRS.3SG
 'My house is more sellable than your house.'
- b. MAX-d: $\mu_c(\text{sell}(\text{your-house}))(d) > \text{MAX-d: } \mu_c(\text{sell}(\text{my-house}))(d)$



Importantly, it is not the POS morpheme that captures the degree variable in the comparative structures, but *daa* 'more'. Given that POS is blocked in them, speakers find it odd to use the comparative structure when they intend to express the SHOULD reading, as shown by the examples in (28).

- (28) a. #*Şk'imi livadi sk'ani livadi-şa daa o-tzil-oni on.*
 my garden your garden-ABL more PV-harvest-ONI be.PRS.3SG
 Int. 'My garden is more in need of harvesting than your garden.'
 (Context: Both of us have gardens with tea plants ready for harvesting. However, unlike yours, mine has been exposed to more sunlight and less rain and it is urgent that the tea in my garden gets harvested. Otherwise, I won't be able to sell it.)
- b. #*Ma sk'ani-şa daa laci o-şv-oni vore.*
 I your-ABL more medicine PV-drink-ONI be.PRS.1SG
 Int. 'I am more in need of taking the medicine than you.'
 (Context: Both of us are ill but I am in a worse state than you and there is only one pill left at home.)

Finally, analyzing ONI as a gradable modal whose COULD reading requires default existential closure within the VP domain (Diesing, 1992) predicts that COULD reading will disappear when the ONI-clause is outside the VP (Grano, 2024). Known as the MODAL DETERMINER GENERALIZATION (Hackl and Nissenbaum, 2012; Bhatt, 1999), the COULD reading is predicted to be unavailable when the modal occurs in the restrictor of a strong determiner, given the strong DPs have to vacate the VP (Diesing, 1992). This is borne out as shown in (29).

- (29) a. *K'at'a gama-şam-oni oxori ham çoyi-s on.*
 every PV-sell-ONI house this village-LOC be.PRS.3SG
 'Every house that should/#can be sold is in this village.'
- b. *Ham çoyi-s gama-şam-oni oxorepe on.*
 this village-LOC PV-sell-ONI house.PL be.PRS.3SG
 'In this village, there are houses that should/can be sold.'

For concreteness, we assume that ONI-clauses in attributive position involves a raising derivation where the abstraction is due to the head noun moving out. Hence, attributive ONI-clauses end up denoting functions of type $\langle d, t \rangle$, which are able to combine with POS. As shown in (30), the idea here is that default existential closure (as opposed to using POS) will only be available if this predicate remains within the VP, otherwise requiring POS.

- (30)
- $$\begin{array}{c}
 \lambda x. \text{house}(x) \wedge \text{MAX-d: } \mu_c(\text{sell}(x))(d) > d_c \\
 \swarrow \quad \searrow \\
 \lambda x. \text{MAX-d: } \mu_c(\text{sell}(x))(d) > d_c \quad \text{oxori} \\
 \swarrow \quad \searrow \\
 \text{MAX-d: } \mu_c(\text{sell}(g(5)))(d) > d_c \quad \lambda_5 \\
 \swarrow \quad \searrow \\
 \text{POS} \quad \lambda d. \mu_c(\text{sell}(g(5)))(d) \\
 \swarrow \quad \searrow \\
 t_5 \quad \lambda x. \lambda d. \mu_c(\text{sell}(x))(d) \\
 \quad \text{gamamşamoni}
 \end{array}$$

All in all, we have presented evidence that ONI contributes a degree modal, correctly predicting its force variability as well as the syntactic restrictions on it. This brings further support to the analysis of force variability developed in Grano (2024) for *for*-infinitivals in English.

4 Structural Composition of ONI-clauses

In this section, we address the question of what the internal syntax/composition of ONI-clauses is like. We start by noting that the subject of predicative ONI-clauses can be understood as the agent or the theme of the event described in the ONI-clause. Notably, this is so regardless of whether the verbal root that the ONI-clause is based on is transitive. As seen in (31c), when the subject is a theme NP, there is no passive morphology of any kind appearing in the ONI-clause.

- (31) a. *Bere-k puci nç'val-um-s*
 child-ERG cow milk-IMPF-PRS.3SG
 'The child is milking the cow.' finite transitive verb
- b. *Bere [puci o-nç'val-oni] on.*
 child cow PV-milk-ONI be.PRS.3SG
 'The child could/should milk the cow.' predicative ONI, agent subject
- c. *Puci [o-nç'val-oni] on.*
 cow PV-milk-ONI be.PRS.3SG
 'The cow could/should be milked.' predicative ONI, theme subject

In Section 3, we have suppressed the composition of ONI-clauses and directly taken verbal roots suffixed with ONI to have a denotation along the lines of (32). We slightly modify the entry here so that the individual is both a participant of the event being described and serves as an anchor in building the modal base, which we will argue is justified independently.

- (32) $\llbracket P - oni \rrbracket^{w,c} = \lambda x. \lambda d. \mu_c(P(x))(x)(d)$
 'for any Q of type $\langle e, vt \rangle$ and any individual y:
 $\lambda d. \mu_c(Q(y))(y)(d)$ characterizes the set of degrees d' returned by a contextually provided measure function μ_c such that it is d'-preferred/permissible for there to be an event e in a world w' such that $Q(y)(e)=1$ and w' is compatible with the circumstances of y at w.'⁹

If this is on the right track, we infer that ONI denotes a function of type $\langle \langle e, vt \rangle, \langle e, dt \rangle \rangle$, predicting that it can take verbs that require a theme complement as well as unsaturated voicePs as complements, as schematized in (33).

⁹We assume that μ_c would be responsible for distinguishing teleological, bouletic, and deontic prioritization.

- (33) a. $\lambda x. \lambda d. \mu_c(\llbracket n\check{\varsigma}'val \rrbracket(x))(x)(d)$
- $\begin{array}{cc} & \diagup \quad \diagdown \\ V_{\langle e, vt \rangle} & \text{oni} \\ n\check{\varsigma}'val & \end{array}$
- b. $\lambda x. \lambda d. \mu_c(\llbracket \text{VOICE } puci \ n\check{\varsigma}'val \rrbracket(x))(x)(d)$
- $\begin{array}{cc} & \diagup \quad \diagdown \\ \lambda y. \lambda e. \text{ag}(e)=y \wedge \llbracket puci \ n\check{\varsigma}'val \rrbracket(e) & \text{oni} \\ \diagup \quad \diagdown & \\ \text{VOICE} & \text{VP}_{\langle v, t \rangle} \\ & puci \ n\check{\varsigma}'val \end{array}$

Even though this is an attractive option, we will argue that the structure in (33b) is not tenable (whereas the structure in (33a) can be maintained). We will present a number of facts suggesting that the complement of ONI never contains an agentive voice layer in the sense of Kratzer (1996). Hence, unlike what previous work has established for finite clauses in Laz (Taylan and Öztürk, 2014; Öztürk and Taylan, 2017; Öztürk, 2021), we argue that ONI-clauses are akin to restructuring complements (Wurmbrand, 2001).

First, there is no morphological evidence for VOICE. In Laz, for unaccusative roots such as *ğur* ‘die’ to license an agent, the causativizing suffix *-in* is required in control clauses and finite clauses (Demirok, 2022; Demirok and Öztürk, 2024) while causative morphology is illicit in ONI-clauses, as shown in (34).

- (34) a. *Bere* [*tzitzila o- ğur -(*)in -oni*] *on*
 child snake PV- die -CAUS -ONI be.PRS.3SG
 ‘The child should/could kill the snake.’ ONI-clause
- b. *Bere-k* [*PRO tzitzila o- ğur -(*)in -u*] *gor-um-s*
 child-ERG PRO snake PV- die -CAUS -NMLZ want-IMPF-PRS.3SG
 ‘The child wants to kill the snake.’ control clause
- c. *Bere-k tzitzila o- ğur -(*)in -am -s*
 child-ERG snake PV- die -CAUS -IMPF -PRS.3SG
 ‘The child is killing the snake.’ finite clause

Second, instrumental adjuncts are *not* licensed in ONI-clauses while they are licensed in canonical control complements as well as finite clauses, as illustrated in (35).

- (35) a. *Bere* [*(*)xami-te cari o-k’vat-oni*] *on*
 child knife-INST bread PV-cut-ONI be.PRS.3SG
 ‘The child should cut bread *(with the knife).’ ONI-clause
- b. *Bere-k* [*PRO xami-te cari o-k’vat-u*] *gor-um-s*
 child-ERG PRO knife-INST bread PV-cut-NMLZ want-IMPF-PRS.3SG
 ‘The child wants to cut the bread with the knife.’ control clause

- c. *Bere-k xami-te cari k'vat-um-s*
 child-ERG knife-INST bread cut-IMPF-PRS.3SG
 'The child is cutting the bread with the knife.' finite clause

Third, the subject that the predicative ONI-clause combines with does not have to be an agent (or an NP with a semantic role suitable for an external argument). This is so even when the ONI-marked verb is built on a transitive verbal root whose subject would be required to be an agent external argument. To illustrate, the sentence in (36) seems to have an interpretation where the circumstances of K'va Livadi is what requires tea-picking to happen (in it).

- (36) *K'va Livadi [nçai o-tzil-oni] on*
 rock garden tea PV-pick-ONI be.PRS.3SG
 ≈ 'K'va Livadi needs tea-harvesting.'
 (Given the circumstances of K'va Livadi, there should be tea-harvesting in it.).

Compare this to the canonical finite sentence in (37) built using the same root, where the subject has to be an agent.

- (37) a. *Xordza-k K'va Livadi-s nçai tzil-um-s*
 woman-ERG rock garden-LOC tea pick-IMPF-PRS.3SG
 'The woman is picking tea in K'va Livadi.'
- b. **K'va Livadi-k nçai tzil-um-s*
 rock garden-ERG tea pick-IMPF-PRS.3SG
 Int. 'Tea-picking is happening in K'va Livadi.'

Fourth, high applicatives that are canonically taken to be introduced below voice are also not licensed in ONI-clauses, implying the absence of the projections above it. Laz has high applicatives which, in finite clauses, introduce a dative NP as well as an applicative prefix on the verb, as shown in (38) (Öztürk, 2013; Öztürk, 2019).

- (38) *Ma himu-s cari v-u-ç'op-i*
 I 3SG-DAT bread 1-APPL.3SG-buy-PST
 'I bought bread for him.'

In control clauses, high applicatives are still licensed while the applicative morphology cannot surface on the embedded nominalized verb (Demirok and Öztürk, 2024), as shown in (39a). In ONI-clauses, high applicatives are banned altogether, as shown in (39b).

- (39) a. *himu-s cari o-(*)-ç'op-u b-gor-um*
 3SG-DAT bread PV-APPL.3SG-buy-NMLZ 1-want-IMPF
 'I want to buy bread for him.'
- b. *(*)Himu-s cari o-(*)-ç'op-oni vore*
 3SG-DAT bread PV-APPL.3SG-buy-ONI be.PRS.1SG
 'I should buy bread (*for him).'

Notably, there is no difficulty in licensing low applicatives in ONI-clauses, as shown in (40).

- (40) a. *Himu-s para o-ncğon-oni vore*
 3SG-DAT money PV-send-ONI be.PRS.1SG
 ‘I should send money to him.’ ONI-clause
- b. *Ma [PRO himu-s para o-ncğon-u] b-gor-um*
 I PRO 3SG-DAT money PV-send-NMLZ 1-want-IMPF
 ‘I want to send money to him.’ control clause

Fifth, attributive ONI-clauses cannot modify an NP that would have an agent role in the event.

- (41) a. *o-nç’val-oni pucepe*
 PV-milk-ONI cow.PL
 ‘cows that can be milked’
- b. **puci o-nç’val-oni k’oçepe*
 cow PV-milk-ONI man.PL
 Int. ‘men that can milk the cow’

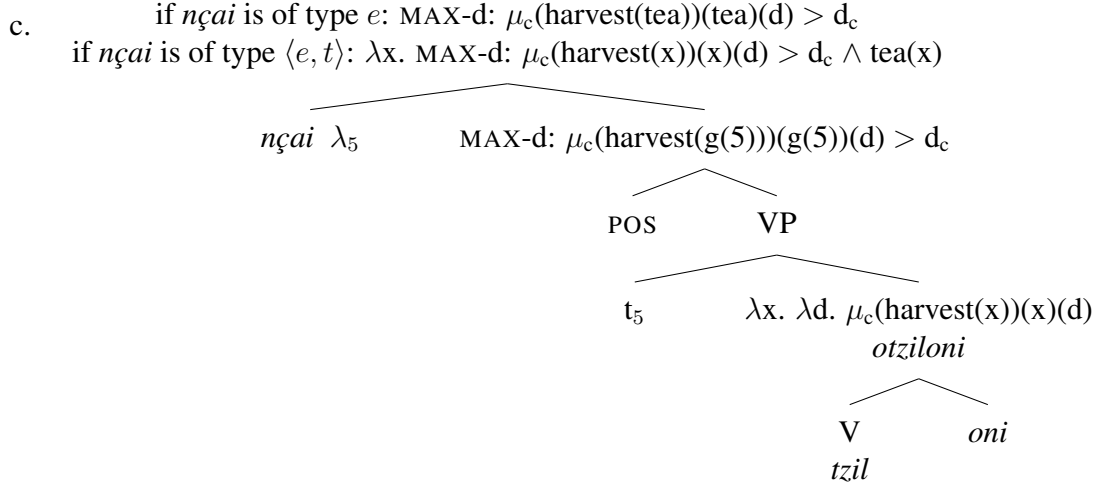
These facts all point towards the conclusion that there is no position for an agent within the ONI-clause. Therefore, we propose that the subject is used to build the modal base. To illustrate, in the sentence in (42), it is the circumstances of the subject *t’aroni* ‘the weather’ that is relevant in making a modal claim about a swimming event.

- (42) *T’aroni [o-nçir-oni] on.*
 weather PV-swim-ONI be.PRS.3SG
 ≈ ‘Weather conditions are suitable for swimming.’

Given that there is no evidence for voice being part of the complement of ONI, we will assume that the complement of ONI is not uniformly of type $\langle e, vt \rangle$. We assume that there are two types of ONI entries, one that combines with VPs and one that combines with Vs. The latter whose denotation is given in (43) is employed in attributive positions and in predicative constructions where the subject is a theme NP, as demonstrated in (44).

$$(43) \quad \llbracket \text{oni} \rrbracket_V = \lambda Q_{\langle e, vt \rangle}. \lambda x. \lambda d. \mu_c(Q(x))(x)(d)$$

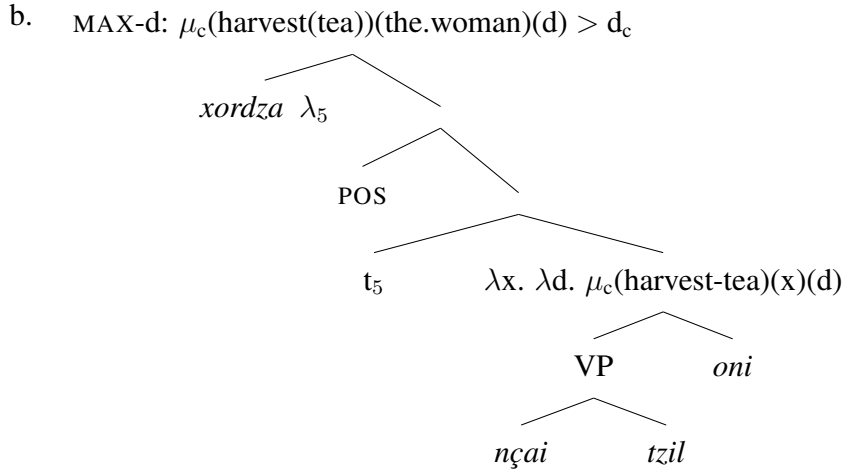
- (44) a. *nçai [o-tzil-oni] on*
 tea PV-pick-ONI be.PRS.3SG
 ‘The tea should be harvested.’
- b. *[o-tzil-oni] nçai*
 PV-pick-ONI tea
 ‘tea which should be harvested’



Now, we turn to the structures where the complement of ONI is a VP, hence of type $\langle v, t \rangle$. The denotation of ONI to be used in such configurations is given in (45).

$$(45) \quad \llbracket \text{oni} \rrbracket_{VP} = \lambda P_{\langle v, t \rangle}. \lambda x. \lambda d. \mu_c(P)(x)(d)$$

- (46) a. *Xordza* [*nçai o-tzil-oni*] *on*
 Woman tea PV-pick-ONI be.PRS.3SG
 ‘The woman should harvest tea.’



We speculate that what blocks the formation of an attributive ONI-clause using the VP-selecting ONI is syntactic in nature. Only ONI that directly combines with a V is able to produce a modifier akin to an adjective. In that sense, V-selecting ONI can be seen as a derivational suffix deriving adjectives from verbs, unlike VP-selecting ONI, which cannot do so.

We end this section with a piece of evidence for the proposed structural asymmetry. The evidence comes from low applicative structures being impossible in cases where a theme NP is the

subject. Under our analysis, the configuration in (47a) is arguably ruled out because ONI directly combines with the V, providing no position for the applicative head. However, low applicatives are licensed when the subject is not a theme NP, as shown in (47b). This is again predicted because the complement of ONI is a VP in this case, rather than V.

- (47) a. **K'argo* [*xordza-s* *o-ncğon-oni*] *on*.
 cargo woman-DAT PV-send-ONI be.PRS.3SG
 Int. 'The cargo should be sent to the woman.'
- b. *Ma* [*xordza-s* *k'argo o-ncğon-oni*] *vore*.
 I woman-DAT cargo PV-send-ONI be.PRS.1SG
 'I should send (the) cargo to the woman.'

5 Concluding remarks

We have argued that the force variability of the modal suffix *-oni* in Laz can be best understood in terms of a degree-based semantics, following the analysis in Grano (2024). We have documented compelling evidence for this approach, in the form of restrictions on force variability that surface in certain syntactic contexts such as being in the restrictor of a strong determiner and serving as a gradable predicate in comparative structures.

Our investigation has also revealed that the subject of a root modal could serve as an anchor for building the modal base without necessarily being asserted as a participant of the event. While we do believe that the subject of ONI-clauses helps build a modal base, it remains unclear if its link to the event is to be established in the semantics. While there is no evidence for an agentive voice layer in the syntax of an ONI-clause, it might as well be the case that VP-selecting ONI in addition links the subject to a contextually suitable thematic role such as agent or location and relates it to the event as a participant. This can be achieved by modifying the denotation of VP-selecting ONI as in (48).

$$(48) \quad \llbracket oni \rrbracket_{VP} = \lambda P_{\langle v, t \rangle}. \lambda x. \lambda d. \mu_c(\lambda e. P(e) \wedge \theta_c(e)=x)(x)(d)$$

Finally, in our compositional analysis, we have assumed that the denotations of predicative ONI-clauses with theme subjects and attributive ONI-clauses are identical. They both involve the V-selecting ONI. An alternative analysis could be assuming that VP-selecting ONI is used for all predicative ONI-clauses. Under this analysis, the theme argument, appearing as the subject, would need to bind a PRO in the thematic object slot in the ONI-clause. Accordingly, the low applicatives not being possible could be explained as an intervention in the licensing of the PRO theme in the ONI-clause. We have not found decisive evidence differentiating the two analyses.

We hope that this study has not only helped document a novel case of a variable-force modal in an endangered language but also brought further support to the notion that some variable-force modals may need degree-based semantic analyses.

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